Small-Scale Proxies for Large-Scale Transformer Training Instabilities

Report — MVA 2025–2026

Tristan Beruard, Fotios Kapotos
CentraleSupélec / MVA
tristanberuard@gmail.com, fotiskapotos@gmail.com

Abstract

We reproduce and investigate two training instabilities described in Wortsman et al. [2023]: *attention logit growth* and *output logit divergence*. We show that both phenomena can be elicited in small Transformers (under 10 M parameters) by training at high learning rates, and that the mitigations proposed by the authors—**qk-layernorm** and **z-loss** regularization—are equally effective at small scale. Our experiments validate the paper's central claim that small models serve as reliable proxies for studying large-scale instabilities, without requiring massive compute.

1 Introduction

Scaling Transformer-based models has driven much of the recent progress in deep learning, from large language models to vision systems. Yet larger models do not always train smoothly: practitioners have repeatedly reported training instabilities that only appear at scale, making them costly and difficult to study Chowdhery et al. [2022], Dehghani et al. [2023]. A team at a large lab may only discover a failure mode after burning weeks of compute on a multi-billion-parameter run, with little insight into the root cause.

Wortsman et al. [2023] propose a practical remedy: *small models can serve as proxies for large-scale instabilities*. Their key observation is that the same failure modes that appear in models with tens of billions of parameters can be reproduced in models with just a few million parameters, simply by training at unusually high learning rates. This makes the instabilities cheap to study, ablate, and fix.

The paper identifies two concrete instabilities and their mitigations:

- **Attention logit growth** Dehghani et al. [2023]: the queries and keys in self-attention layers grow unboundedly, causing the softmax to collapse to a one-hot distribution and learning to stall. The proposed fix is *qk-layernorm*, which applies LayerNorm to queries and keys before the dot product.
- **Output logit divergence** Chowdhery et al. [2022]: the output logits drift to very negative values, decoupling them from the log probabilities. The proposed fix is an auxiliary *z-loss* term that penalises $\log^2 Z$, keeping the softmax partition function near 1.

To summarise the sensitivity of a model to learning rate across all these conditions, the authors introduce a scalar metric called *LR sensitivity*, which measures the expected deviation from optimal loss when sweeping the learning rate over three orders of magnitude.

[WIP: Clearly write what we did]

Our contribution. In this report we independently reproduce both instabilities at small scale and verify that the proposed mitigations work as claimed. Concretely, we train decoder-only Transformers with fewer than 10 M parameters on the **[TODO: what dataset, FineWeb probs ?]**, sweep

Preprint. Under review.

learning rates from 3×10^{-4} to 3×10^{-1}, and confirm that (i) attention logit growth and its characteristic loss divergence appear at high learning rates without qk-layernorm, and (ii) output logit divergence appears without weight decay, and is reliably cured by z-loss. Our experiments require only modest compute and constitute a self-contained verification of the paper's central claim.

2 Paper Summary

Wortsman et al. argue that large-scale Transformer training instabilities are not caused by scale per se, but by the interaction between architecture, optimizer, and learning rate. As a result, the same failure modes can be elicited in small models at high learning rates, making them cheap to study and fix. To quantify this, the authors introduce *LR sensitivity*, defined as:

$$\mathcal{S} = \mathbb{E}_{\eta \in [a,b]} \left[\min\big(\ell(\mathcal{A}(\eta)),\, \ell_0\big) - \ell^* \right], \tag{1}$$

where $\ell(\mathcal{A}(\eta))$ is the validation loss when training with learning rate η, $\ell^* = \min_\eta \ell(\mathcal{A}(\eta))$ is the best achievable loss over the swept range, and ℓ_0 is the loss at initialisation (which caps diverged runs). A high $\mathcal{S}$ means the model requires careful LR tuning and is likely to diverge if η is even slightly too large.

Attention logit growth. In self-attention, the logits $z_{ij} = \langle q_i, k_j \rangle / \sqrt{d_h}$ depend on the *product* of query and key vectors. Because both q_i and k_j grow linearly with their respective weight matrix norms, the logit magnitudes grow *quadratically* with parameter norm — much faster than any other feature in the network. Once logits become very large, the softmax saturates into a near-one-hot distribution: every token attends to a single fixed position regardless of content, and the model stops learning. First reported in a 22 B parameter ViT Dehghani et al. [2023], the paper shows this also occurs in 9 M parameter models at high LR. The mitigation is **qk-layernorm**: apply LayerNorm to q and k before the dot product, which pins their norms and breaks the quadratic dependence.

Output logit divergence. The model's output logits $y \in \mathbb{R}^V$ produce token probabilities $p_i = e^{y_i}/Z$ where $Z = \sum_j e^{y_j}$. Without weight decay, the logits drift collectively towards very negative values late in training. As $y_i \to -\infty$, $Z \to 0$ and the log-probabilities $\log p_i = y_i - \log Z$ decouple from the logits, causing the loss to stall. First reported in PaLM Chowdhery et al. [2022], the paper reproduces it in a 2.4 M parameter model. The mitigation is a **z-loss** auxiliary penalty $\alpha \log^2 Z$ (with $\alpha = 10^{-4}$), which softly keeps $\log Z \approx 0$ throughout training.

3 Claims Under Verification

[WIP: This is not yet definite, to be changed when experiments are done] **Claim 1.** *Attention logit growth appears in small Transformers at high learning rates, and qk-layernorm resolves it at both small and large scale, measurably reducing LR sensitivity.* We verify this by training models with and without qk-layernorm across several learning rates, tracking the max attention logit over training, and computing LR sensitivity for each configuration (Section 3.1.1 and Figures 1–2 of Wortsman et al. [2023]).

Claim 2. *Output logit divergence occurs in models of any size when weight decay is absent, and z-loss reliably prevents it.* We verify this by training the four combinations of {z-loss, no z-loss} $\times$ {weight decay, no weight decay} and monitoring the output logit mean throughout training (Section 3.1.2 and Figures 3–4 of Wortsman et al. [2023]).

References

A. Chowdhery et al. PaLM: Scaling language modeling with pathways. *arXiv preprint arXiv:2204.02311*, 2022.

M. Dehghani et al. Scaling vision transformers to 22 billion parameters. 2023.

M. Wortsman, P. J. Liu, L. Xiao, K. Everett, A. Alemi, B. Adlam, J. D. Co-Reyes, I. Gur, A. Kumar, R. Novak, J. Pennington, J. Sohl-Dickstein, K. Xu, J. Lee, J. Gilmer, and S. Kornblith. Small-scale proxies for large-scale Transformer training instabilities. *arXiv preprint arXiv:2309.14322*, 2023.